

Exercise 1

- Consider a one-qubit system in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle - e^{i\theta} |1\rangle)$$

- Consider the observable

$$A = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

- Find the expectation value of A , for what value of θ is it minimized?

Exercise 1

- Find the probability distribution of A . Use this result to confirm your calculation of the expectation value
- [Challenge] After you measured A , you measure another observable

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

- Imagine you repeat this experiment many times. What is the probability distribution of your measurement of B ?

Exercise 2

- Consider a two-qubit system. Suppose we have two operators (expressed in the computational basis) acting on the two qubits respectively:

$$A_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- What are the possible outcomes of measuring the following operator on the total, composite system?

$$A_1 \otimes A_2$$